

EB-NeRD Large

Assignment 2 Design Note

Two-stage news recommendation with behavioural features and CatBoost learning-to-rank

Abstract

This design note describes the EB-NeRD-only implementation for Assignment 2. The system extends the Assignment 1 lexical/semantic candidate pipeline with click-history, freshness, popularity, category, session, and position features, followed by a CatBoost PairLogit re-ranker. A frozen 768-dimensional EB-NeRD contrastive embedding artifact is used for semantic history features. The principal evaluation uses 60,000 impressions from the second half of the validation week under a stale, train-only behavioural regime. CatBoost improves the A1 reference from AUC 0.68060 to 0.71287 and improves MRR, nDCG@5 and nDCG@10; paired impression-level bootstrap 95% confidence intervals exclude zero for all four gains. An NRMS-style frozen-document-vector experiment is also reported as an alternative neural ranker. Serving measurements, scaling limits, feature-masking diagnostics, and the complete 13.5M-impression test prediction pipeline are documented.

1. Scope and Objectives

The Assignment 2 specification asks the team to extend the Assignment 1 lexical/semantic retrieval system with behavioural signals, implement a two-stage retrieve-then-rank pipeline, report baseline versus improved ranking metrics, perform an ablation with paired bootstrap confidence intervals, analyse serving and 10× scale, and report beyond-accuracy slices. This document covers **EB-NeRD only** and records the executed experiments and their limitations rather than substituting unmeasured results.

The final production-style experiment is a CatBoost learning-to-rank re-ranker. The A1 hybrid remains the reference system. A separate NRMSDocVec experiment is retained as an alternative neural-ranker result; it is not used for the final EB-NeRD submission.

2. System Overview

Offline: split the data temporally; construct split-specific user histories; build freshness, popularity and decayed-CTR features; compute semantic/BM25/category history features; train the ranker on impression groups.

Online conceptual path: A1 provides the candidate slate; serving-time features are computed from the current user history and article metadata; CatBoost scores each candidate; scores are sorted within each impression and emitted as an ordinal permutation.

Stage	EB-NeRD implementation
Corpus	125,541 articles; large train/validation/test behaviour sets
Candidate/reference	A1 hybrid lexical + semantic/freshness/popularity reference
Re-ranker	CatBoostRanker, PairLogit, 700 iterations, depth 8
History	Most recent 30 clicks; split-specific 21-day lookbacks
Validation	60,000 impressions from second half of validation
Final submission	13,536,710 test impressions; predictions.txt inside ZIP

The assignment's target deliverables and evaluation requirements are taken directly from the CS4.406 A2 specification.

3. Dataset, Temporal Protocol and Data Integrity

EB-NeRD is a large Danish news recommendation dataset. The executed pipeline uses the large bundle and the separate large test set. Only the columns needed by the ranking pipeline are projected from Parquet, which is important because the raw behaviours contain additional post-impression and demographic fields.

Split	Impressions	Time interval	History used
Train	12,063,890	2023-05-18 07:00 → 2023-05-25 06:59:59	train history
Validation	12,566,385	2023-05-25 07:00 → 2023-06-01 06:59:59	validation history
Test	13,536,710	2023-06-01 07:00 → 2023-06-08 06:59:59	test history
Articles	125,541	catalogue	supplied contrastive vectors

The test set contains 205,925,868 in-view candidate rows, averaging 15.2 candidates per impression. The supplied contrastive artifact contains a 768-dimensional vector for every catalogue article; after normalization the vectors occupy approximately 386 MB as float32.

3.1 Behaviour-window boundary

The implementation keeps train, validation and test histories as separate objects. The history parquet is read as the most recent 30 article IDs per user. Split boundaries are derived directly from the corresponding behaviour table: train starts 2023-05-18 07:00, validation 2023-05-25 07:00, and test 2023-06-01 07:00. An executable assertion confirms that the three histories remain separate and are not mixed.

The supplied history object used by the final scorer stores user-to-clicked-article lists; the boundary test therefore verifies split-specific history sourcing rather than reconstructing a timestamp inequality for every individual click. This distinction is retained explicitly to avoid overstating the test.

3.2 Causal session context

EB-NeRD exposes `session_id`. A zero-based session-depth feature is computed after sorting each split by impression time and independently assigning positions within each session. Thus the depth of an impression depends only on earlier rows in that session. **read_time** and **scroll_percentage** are excluded because they are post-impression outcomes and are not known when the recommendation is served.

3.3 Label construction and evaluation protocol

For labelled impressions, an in-view candidate receives label 1 when its article ID occurs in `article_ids_clicked` and 0 otherwise. Metrics are computed per impression and then aggregated across valid slates. A fast vectorized evaluator was checked against the official metric definitions; the recorded validation comparison showed identical AUC and only sub- 1.3×10^{-4} absolute differences for the rank metrics in the implementation check.

4. Feature Engineering

The feature set combines the established A1 ranking signal with behavioural, semantic, article and session features. The design intentionally separates article-constant arrays from user-history-derived batch computations so that the large test pass remains memory-bounded.

Feature	Construction	Serving status
a1_score	A1 hybrid score combining retrieval/ranking signals	Reference-stage signal
log_pageviews	$\log(1 + \text{total_pageviews})$	Unavailable / future-contaminated
log_inviews	$\log(1 + \text{total_inviews})$	Dataset-provided; availability depends on deployment
decayed CTR	24 h half-life; Bayesian smoothing $\alpha=20$	Historical-only table
age_bucket	10 publication-age buckets	Serving-safe from publication time
semantic_history	$\text{cosine}(\text{candidate vector}, \text{normalized history vector})$	Serving-safe
bm25_history	BM25 candidate score against history-derived query	Serving-safe
category_affinity	candidate category frequency in user history	Serving-safe
subcategory_affinity	candidate subcategory frequency in user history	Serving-safe
sentiment	article sentiment score	Article metadata
position_norm	$\text{candidate position} / (\text{slate length} - 1)$	Serving-safe
session_depth	causal zero-based impression position in session	Serving-safe
history_len	number of retained historical clicks	Serving-safe

4.1 Recency and popularity

The article-age function is bucketed rather than linear. On the large training logs, the measured click-rate peak occurs in the 8–12 h bucket, with a $9.4\times$ max/min spread across the ten buckets. The training-derived values are: **<1h 0.0838; 1–2h 0.1161; 2–4h 0.1263; 4–8h 0.1292; 8–12h 0.1319; 12–18h 0.0975; 18–24h 0.0497; 24–48h 0.0334; 48–120h 0.0288; >120h 0.0141.**

A decayed article CTR is accumulated from labelled impressions using a 24-hour half-life, with Bayesian smoothing toward the global rate ($\alpha=20$). Three temporal versions are kept: FRESH includes the first half of validation, STALE uses train only, and FINAL uses all labels strictly before the test week. The main CatBoost validation uses STALE so that its temporal information matches the train-only regime expected before test-time labels.

4.2 History representation

For each user, the last 30 clicks form a sparse user $\times$ article matrix. Semantic history is obtained by sum-pooling the user's article vectors and L2-normalizing the result. BM25 history uses the same sparse click matrix to form a history-derived term-count query. Category and subcategory affinities are normalized historical frequencies.

4.3 Article and position features

The supplied contrastive vectors cover all 125,541 articles. Candidate position is normalized within each impression. Position is retained as a behavioural/ranking-context feature, while post-impression fields are deliberately excluded.

5. Ranking Models and Experimental Design

5.1 A1 reference and CatBoost re-ranker

The A1 hybrid is the reference ranking system. Its stale validation blend uses the tuned weights **pv=3.0, ctr=0.5, age=3.0, sem=1.0, bm25=0.5, sub=0.5**. A no-pageviews A1 diagnostic uses pv=0, ctr=1, age=3, sem=1, bm25=0.5, sub=0.5. The best full blend reached AUC 0.6806 on the 60K validation subset.

CatBoost is trained as a grouped learning-to-rank model using PairLogit. Training uses 300,000 sampled train impressions, producing 3,324,782 candidate rows and 301,064 positive labels. Configuration: 700 iterations, depth 8, learning rate 0.08, L2 leaf regularization 5, random strength 1, and CPU parallelism. The resulting model file is approximately 2.95 MB.

Model	AUC	MRR	nDCG@5	nDCG@10
A1 reference	0.68060	0.45311	0.50802	0.55911
CatBoost PairLogit	0.71287	0.47789	0.54361	0.58548
CatBoost – A1	+0.03227	+0.02478	+0.03559	+0.02637

The CatBoost improvement is consistent across all four metrics. Relative to A1, the absolute gains are +0.03227 AUC, +0.02478 MRR, +0.03559 nDCG@5 and +0.02637 nDCG@10.

5.2 NRMSDocVec alternative

A separate NRMS-style experiment was trained using the supplied 768-dimensional contrastive vectors as frozen article inputs. The trainable network projects 768→512→256, applies a 16-head self-attention user encoder followed by additive attention, and scores candidates by user–article dot product. It used 1,000,000 sampled train impressions, 8 negatives, batch size 256, 8 epochs and learning rate 10^{-4} . Filtering yielded 492,151 usable training examples. The model had approximately 0.84M trainable parameters.

Model	AUC	MRR	nDCG@5	nDCG@10
A1 reference	0.68060	0.45311	0.50802	0.55911
NRMSDocVec	0.53547	0.33726	0.36967	0.45201
NRMS – A1	–0.14513	–0.11585	–0.13835	–0.10710

NRMSDocVec therefore did not reproduce the A1 reference performance on this run. This result is retained as an experimental comparison; the final EB-NeRD test submission uses CatBoost.

5.3 Why CatBoost is the final re-ranker

The CatBoost model can jointly exploit heterogeneous scalar signals (freshness, popularity, CTR, position and session context) and user-history-derived signals. The executed validation results show a larger improvement over A1 than the frozen-vector NRMS experiment. This is an empirical observation for this implementation and validation protocol, not a general claim that GBDT models dominate neural rankers.

6. Ablation, Statistical Significance and Extended Evaluation

6.1 Paired bootstrap against A1

Statistical uncertainty is estimated using 1,000 paired bootstrap resamples at the impression level. Both systems are evaluated on the same impression and candidate slate in each resample. The reported delta is CatBoost minus A1.

Metric	Δ	95% CI
AUC	+0.032269	[0.030164, 0.034350]
MRR	+0.024783	[0.022108, 0.027611]
nDCG@5	+0.035595	[0.032976, 0.038386]
nDCG@10	+0.026372	[0.024105, 0.028673]

All four confidence intervals are strictly positive, so the observed CatBoost gains are separated from zero under this paired-bootstrap procedure.

6.2 History ablation

To isolate the contribution of the personalization path without another expensive training run, the same trained CatBoost model is evaluated after masking semantic_history, bm25_history, category_affinity, subcategory_affinity and history_len. This is explicitly an **inference masking ablation**, not a separately retrained model.

Metric	Full – history-masked	95% CI
AUC	+0.006989	[0.005715, 0.008220]
MRR	+0.010819	[0.009125, 0.012655]
nDCG@5	+0.008624	[0.006981, 0.010184]
nDCG@10	+0.008393	[0.007024, 0.009776]

The positive deltas indicate that history-derived features contribute measurable ranking information beyond the remaining features. The MRR and nDCG changes are larger than the AUC change, consistent with a contribution concentrated in within-slate ordering.

6.3 Serving-unavailable pageviews diagnostic (Q9)

total_pageviews is retained in the leaderboard model because it is part of the available dataset signal, but its definition counts clicks during the first seven days after article publication. For an article published during the test week, this can contain information from after the recommendation timestamp. It is therefore treated as serving-unavailable/future-contaminated for the anti-gaming analysis.

Metric	Full – pageviews-masked	95% CI
AUC	+0.040109	[0.038225, 0.042180]
MRR	+0.045196	[0.042630, 0.047817]
nDCG@5	+0.052818	[0.050496, 0.055279]
nDCG@10	+0.039396	[0.037228, 0.041436]

Masking pageviews produces a large drop, demonstrating substantial dependence of the trained model on this feature. Because this is masking rather than retraining, it should not be described as the performance of an independently trained serving-safe model.

6.4 Beyond-accuracy and slices

Measure	Result	Interpretation
category diversity@10	0.7761	Category-based diversity of top-10 lists
novelty@10	37.0729	Information-content novelty using training click frequency
coverage@10	0.01943	≈1.94% of 125,541-article catalogue exposed in top-10
Warm	60,000	AUC 0.71287; MRR 0.47789; nDCG@5 0.54361; nDCG@10 0.58548
Head	835	AUC 0.38629; small slice, interpret cautiously
Tail	58,596	AUC 0.72154; MRR 0.48347; nDCG@5 0.55063
Cold	0	No examples met the defined <5-history-click criterion

The head slice is much smaller than the tail slice, so the two should not be interpreted as equally precise estimates. No cold-start metric is reported because the selected 60K evaluation subset contained zero impressions under the implemented cold-start definition.

7. Serving, Scale, Reproducibility and Final Test Pipeline

7.1 Measured serving characteristics

Component	Measured value
CatBoost reranking p50	33.23 ms / impression
CatBoost reranking p95	35.44 ms / impression
CatBoost reranking p99	38.41 ms / impression
CatBoost model size	2.95 MB
Frozen article-vector store	385.66 MB
10× catalogue vector store	3.86 GB

The latency benchmark measures the CatBoost **re-ranking stage** for one impression at a time. Candidate generation and feature construction are inherited from A1 and are not included in the 38.41 ms p99 measurement. Consequently, the number should not be presented as a full end-to-end online-request latency.

7.2 Cost/QPS framing and 10× scaling

A simple latency-implied serial capacity is approximately $1/0.03841 \approx 26$ requests/s for the measured re-ranking stage on the benchmark environment; this is a back-of-the-envelope capacity figure, not a measured saturation throughput. For a deployment with hourly compute price P , the corresponding re-ranking cost per 1,000 serial requests is approximately $P \times 1000 / \text{QPS}$. A concrete monetary figure requires selecting a cloud instance and its price, which was not fixed in the executed experiment.

At 10× catalogue scale, the dense vector store grows from 385.66 MB to approximately 3.86 GB because storage is $O(Nd)$. Tree scoring scales with candidate rows, number of trees and tree depth. The practical scaling pressures identified by the benchmark are therefore feature/candidate generation cost, memory bandwidth and dense-index memory. ANN indexing, dimensionality reduction or product quantization are natural next steps if the candidate corpus or query load exceeds the capacity of exact dense retrieval.

7.3 Final test prediction and submission validation

The final CatBoost model scores the complete EB-NeRD test candidate slate in bounded batches. The test set contains 13,536,710 impressions and 205,925,868 candidate rows. The initial output was subsequently repaired without rescore: the existing rank lists were retained and only the incorrect repeated impression IDs were replaced with the ordered IDs from the test behaviour table. The repaired file passed validation for line count, impression-ID order, slate length and permutation validity.

Submission artifact	Status
predictions.txt	13,536,710 lines
Validation	Passed: IDs, lengths and ordinal permutations
Final ZIP	ebnerd_a2_catboost_predictions.zip
ZIP size	≈229.97 MB
Codabench submission	Ready; leaderboard screenshot to be added after upload

8. Limitations and Reproducibility

The main limitations are: (i) the CatBoost model uses pageviews that are not serving-safe under the dataset's seven-day aggregation definition; (ii) the pageview and history ablations are masking diagnostics rather than retrained counterfactual models; (iii) the cold-start slice has no observations in the selected validation subset; and (iv) the latency benchmark covers the re-ranking stage rather than complete retrieval plus feature construction. These are reported explicitly so that measured results are not overstated.

The pipeline is reproducible from the EB-NeRD large/test bundles and the supplied contrastive embedding artifact. The executed notebook stores model-comparison, bootstrap and summary artifacts. Large datasets, embeddings, model files and ZIP predictions should remain outside Git in accordance with the assignment's repository policy.

9. Conclusion

The EB-NeRD A2 implementation demonstrates a two-stage behavioural re-ranking pipeline in which CatBoost substantially improves the A1 reference on the selected temporal validation set. Freshness is the strongest individual measured signal, while history-derived features provide a statistically distinguishable additional contribution. The pageview diagnostic identifies a major serving-safety limitation that should be addressed in a production deployment.

The final CatBoost test predictions are validated and packaged for Codabench.

References

- [1] CS4.406 Information Retrieval & Extraction, Assignment 2 specification, 2026.
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- [3] Wu et al., “Neural News Recommendation with Multi-Head Self-Attention,” EMNLP-IJCNLP 2019.
- [4] Assignment 1 EB-NeRD design and retrieval pipeline, project artifact.